Review of Matrix Differentiation

Let $x \in \mathbb{R}^p$, i.e., $x = (x_1, x_2, \dots, x_p)^T = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_p \end{pmatrix}$. x is $p \times 1$.

Let $g(x) : \mathbb{R}^p \to \mathbb{R}$ be a scalar-valued function. Then

$$\frac{\partial g}{\partial x} = \begin{pmatrix} \frac{\partial g}{\partial x_1} \\ \frac{\partial g}{\partial x_2} \\ \vdots \\ \frac{\partial g}{\partial x_p} \end{pmatrix} \text{ is } p \times 1.$$

e.g., take $g(x) = a_1 x_1 + a_2 x_2 + \cdots + a_p x_p$ for $a_j \in \mathbb{R}, j = 1, 2, \dots, p$. Then

$$\begin{aligned} \frac{\partial g}{\partial x_j} &= \frac{\partial}{\partial x_j} \left(\sum_{k=1}^{p} a_k x_k \right) = \sum_{k=1}^{p} \frac{\partial}{\partial x_j} (a_k x_k) \\ &= \sum_{k=1}^{p} a_k \frac{\partial x_k}{\partial x_j} = a_j \end{aligned}$$

and

$$\frac{\partial g}{\partial x} = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_p \end{pmatrix} = a.$$

Thus, $\frac{\partial}{\partial x}(a^T x) = a$, and because $a^T x = x^T a$, we have $\frac{\partial}{\partial x}(x^T a) = a$.
Also note that

$$\begin{aligned} \frac{\partial}{\partial x^T}(a^T x) &= \left[\frac{\partial}{\partial x}(a^T x) \right]^T = [a]^T = a^T \\ &= \frac{\partial}{\partial x^T}(x^T a) \text{ since } a^T x = x^T a. \end{aligned}$$

Notation: $\frac{\partial}{\partial x} g(x)$ for $x \in \mathbb{R}^p$ is the gradient of g wrt x written as $\frac{\partial}{\partial x} g(x) = \nabla_x g(x)$.

Now let x, y be vectors, $x \in \mathbb{R}^p$, $y \in \mathbb{R}^r$ with elements of y dependent on x.

Let $x \in \mathbb{R}^p$ and $y(x) : \mathbb{R}^p \to \mathbb{R}^r$ be a vector-valued function.

$\frac{\partial y^T}{\partial x}$ is the $p \times r$ matrix with element (i, j) $\frac{\partial y_j}{\partial x_i}$.

$$\frac{\partial y^T}{\partial x} = \begin{pmatrix} \frac{\partial y_1}{\partial x_1} & \frac{\partial y_2}{\partial x_1} & \cdots & \frac{\partial y_r}{\partial x_1} \\ \frac{\partial y_1}{\partial x_2} & \frac{\partial y_2}{\partial x_2} & \cdots & \frac{\partial y_r}{\partial x_2} \\ \vdots & \vdots & \ddots & \vdots \\ \vdots & & & \frac{\partial y_r}{\partial x_p} \end{pmatrix}$$

Also, $\left(\frac{\partial y^T}{\partial x}\right)^T = \frac{\partial y}{\partial x^T}$.

Note $\frac{\partial x}{\partial x^T} = \frac{\partial x^T}{\partial x} = I_p$ - the p-dimensional identity matrix.

Letting A, B not depend on x, $A_{q\times p}, B_{p\times s}$

$$\frac{\partial}{\partial x^T}(Ax) = A\frac{\partial}{\partial x^T}(x) = A \cdot I_p = A$$

$$\frac{\partial}{\partial x}(x^T B) = \left(\frac{\partial}{\partial x}x^T\right) B = I_p \cdot B = B$$

Consider vectors $u, v \in \mathbb{R}^s$ with elements depending on x. The inner product $u^T v = \sum_{j=1}^{s} u_j v_j \in \mathbb{R}$.

$$\underset{p\times 1}{\frac{\partial}{\partial x}}(u^T v) = \frac{\partial}{\partial x}\left(\sum_{j=1}^{s} u_j v_j\right)$$

$$= \sum_{j=1}^{s}\left[u_j \underset{p\times 1}{\frac{\partial}{\partial x}} v_j + v_j \underset{p\times 1}{\frac{\partial}{\partial x}} u_j\right]$$

$$= \left(\underset{p\times s}{\frac{\partial v^T}{\partial x}}\right) \underset{s\times 1}{u} + \left(\underset{p\times s}{\frac{\partial u^T}{\partial x}}\right) \underset{s\times 1}{v}$$

> **Tip**
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> This is the product rule for vector differentiation: $\frac{\partial}{\partial x}(u^T v) = (\frac{\partial v^T}{\partial x})u + (\frac{\partial u^T}{\partial x})v$. It accounts for the fact that both u and v may depend on x.

Quadratic Form

Let $A_{p\times p}$ be constants (not dependent on x). Then for $x \in \mathbb{R}^p$.

$$\frac{\partial}{\partial x}(x^T A x) = \frac{\partial}{\partial x}\left(x^T \begin{pmatrix} a_{1.}^T x \\ a_{2.}^T x \\ \vdots \\ a_{p.}^T x \end{pmatrix}\right) \quad \text{here } a_{1.} \text{ is the first row of } A$$

$$= \frac{\partial}{\partial x}(x_1(a_{1.}^T x) + x_2(a_{2.}^T x) + \cdots + x_p(a_{p.}^T x))$$

$$= \begin{pmatrix} 2x_1 a_{11} + \sum_{k\neq 1} a_{1k}x_k + x_2 a_{21} + \cdots + x_p a_{p1} \\ x_1 a_{12} + 2x_2 a_{22} + \sum_{k\neq 2} a_{2k}x_k + \cdots + x_p a_{p2} \\ \vdots \\ x_1 a_{1p} + x_2 a_{2p} + \cdots + 2x_p a_{pp} + \sum_{k\neq p} a_{pk}x_k \end{pmatrix}^T$$

Warning

The transpose operator (^{T}) applied to the column vector here is incorrect; the gradient is defined as a column vector, and the subsequent expression $Ax + A^T x$ is a column vector.

$$= Ax + A^T x \in \mathbb{R}^p$$

If $A_{p\times p}$ is symmetric, then $\frac{\partial}{\partial x}(x^T Ax) = 2Ax$.

Tip

If A is symmetric ($A = A^T$), then $Ax + A^T x$ simplifies to $Ax + Ax = 2Ax$. This is a very common identity in optimization and machine learning when taking the gradient of a quadratic form.

Exercises

1.

Let $x \in \mathbb{R}^p$ and let $u(x) = Ax$ and $v(x) = Bx$ be vector-valued functions where $A, B \in \mathbb{R}^{m\times p}$. Compute the gradient of $f(x)$ where

$$f(x) = u(x) \cdot v(x) = u(x)^T v(x)$$

Tip

Note that $f(x) = (Ax)^T(Bx) = x^T A^T Bx$. This is a quadratic form $x^T Mx$ where $M = A^T B$. The gradient of a quadratic form $x^T Mx$ is $(M + M^T)x$.

2.

Let $A \in \mathbb{R}^{p\times p}$ be a real matrix (not necessarily symmetric) and $b \in \mathbb{R}^p$ a constant vector. For $x \in \mathbb{R}^p$, let

$$f(x) = \frac{1}{2}x^T Ax - b^T x$$

Compute the gradient of f, $\nabla f(x)$, and the Hessian matrix, $\nabla^2 f(x)$.

Tip

Use the standard identities for matrix calculus: $\nabla_x(b^T x) = b$ and $\nabla_x(x^T Ax) = (A + A^T)x$. The Hessian is the derivative of the gradient.